CUSTOMER EXECUTION GUIDE

Oracle to Azure Database for PostgreSQL

Assessment runbook

Guided SQL*Plus collection | Optional host evidence
Azure scratch preparation | Microsoft PostgreSQL extension for VS Code

Document control	Value
Version / issue date	1.8 / 29 October 2026
Audience	Customer DBA, host operator, Azure platform owner and assessment operator
Deliverable	CSV evidence pack, conversion project, findings and handover record
Companion package	The self-contained runbook folder; instructions, scripts and output stay together.
Customer / engagement	Complete before use: _____
Assessment lead / approver	Complete before use: _____

READ BEFORE EXECUTING Oracle collection is read-only. Azure provisioning is paid and requires approval. Schema conversion writes to and drops matching schemas in a disposable scratch database. Application conversion edits a copy of source code. This process does not migrate production rows.

This is a customer-neutral guide. Replace every example with approved customer values. No lab IP addresses, credentials or customer-specific configurations are included.

Product UI and model prerequisites were checked against Microsoft Learn on 20 October 2026. Recheck the linked official guidance immediately before execution.

How to use this runbook

Follow Parts A, B and C in that order. Stop at each approval gate if you do not have the required authority. The checklists and working record are included so operators do not need to navigate multiple guidance documents.

PLACEHOLDERS Replace every <ANGLE_BRACKETED_VALUE>, including the brackets, before execution. Script 99 is the single guided Oracle collection entry point.

Part	Do	Keep
A - Collect	Start script 99 once per approved database/PDB, select approved schemas, and complete its guided workload pause. Optional host collection remains separate. An optional offline HTML review can follow (section 11a).	CSV/TXT outputs, errors, workload labels and operator record.
B - Prepare Azure	Prepare a disposable Azure PostgreSQL Flex scratch database and supported Foundry deployment.	Resource IDs, settings, network tests, approval and cost record.
C - Assess	Run VS Code schema conversion; optionally assess application code.	Complete migration project, reports, notes, original/converted files and issues.
D - Handover	Validate, redact, hash, encrypt and transfer the package.	Signed completeness checklist and unresolved evidence gaps.

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1. Scope, safety and approval gates

What you are collecting

The evidence supports a migration architect's later capacity envelope, compatibility assessment and rollout plan. It is not an automatic production SKU recommendation, a production migration or a performance benchmark.

Gate	Required approval / owner	Do not proceed until
G1 - Source evidence	Oracle DBA, application owner and host owner	Database/services, schema allow-list, read access and sampling windows agreed.
G2 - Azure spend and access	Subscription, security and budget owners	Tenant/subscription/region, networking, role assignments and spending approved.
G3 - AI processing	Customer data/IP owner and security	Sending schema definitions and any approved code to Foundry/Copilot is allowed.
G4 - Scratch destruction	Owner of selected PostgreSQL database	Database is disposable; matching schemas and cascade dependencies can be deleted.
G5 - Application copy	Application owner	Only approved copied code is editable; tool/terminal actions are reviewed.

- Do not run AWR/ASH or the optional script 31 as part of this base process. Licensed history requires a separate written licensing and access decision.
- Do not install Statspack, create its snapshots, change schedules or purge history. Only read an existing approved installation.
- Do not run optional source-export script 34 unless separately approved. Extension extraction itself reads definitions; G3 applies even when script 34 is not used.
- Do not start/stop Oracle, change source parameters, grant privileges, open broad firewall access or create paid resources without the customer's explicit approval.
- Never place passwords, API keys, connection secrets, bind values or production rows in Word, scripts, screenshots, evidence logs or chat.

STOP CONDITION If access, scope, business meaning or safety is uncertain, stop that activity and record the gap. A missing measurement is not zero usage. A successful process exit is not evidence completeness.

2. Preflight: people, access and tools

Operator	Responsibility
Oracle DBA	Approve root/PDB or non-CDB scope, privileges, representative periods and error triage.
Host owner	Approve host commands, output location and any elevation; collect on the Oracle host.
Azure administrator	Create or reuse approved scratch resources, configure network/auth/extensions and cost controls.
Assessment operator	Run collection and extension workflow; record versions, files, outcomes and gaps.
Application owner	Approve source-code copy/AI processing; review conversion findings and tests.

Required for Part A

- Customer-installed SQL*Plus and a tested wallet/external credential alias. Complete workstation checks in sections 2a-2c before collection. Python is NOT required for customer collection.
- Approved Oracle read account, database service, version/patch level, environment, host and owner allow-list.
- Existing protected client output folders without spaces. SQL*Plus SPOOL writes on the machine running the client, not automatically on the database host.
- For optional host collection: Bash on the Oracle Linux host, or Windows PowerShell 5.1 / PowerShell 7 on the Oracle Windows host. No package installation is required.

Request source metadata access through the DBA

The collectors need the DBA_* and V$/GV$ views named in their script headers. For extension conversion, the current Microsoft tutorial lists CONNECT, SELECT_CATALOG_ROLE, SELECT ANY DICTIONARY, SELECT on SYS.ARGUMENT$ and SYS.V_$RESOURCE_LIMIT [R1]. These are broad privileges: the DBA/security owner must review and grant the approved access, or provide an approved DBA-run alternative. This document does not authorize grants.

Version boundary - important for Oracle 10g

10G-TARGETED, 21C-TESTED The numbered base sequence was executed on Oracle 21.3, not on 10g. 10.1/10.2 catalog, privilege and client differences still require an approved matching-version lab test. Do not promise 10g compatibility from the 21c result. Extension driver/source-version support is a separate check.

Record every unavailable view/column and whether it affects sizing or compatibility conclusions. Do not silently treat a collector that continues after an Oracle error as a passed collector.

2a. Windows: prepare SQL*Plus

SQL*Plus is the only supported client for Oracle evidence collection in this runbook. Use its terminal application, not a graphical worksheet. The guided runner and recovery commands use this same client throughout.

Install the required Oracle client packages

1. Install an approved SQL*Plus release compatible with the source Oracle database. Confirm the Windows architecture and client/server compatibility with the DBA; testing on one Oracle release does not certify every older source version.

2. For Instant Client, download BOTH the SQL*Plus package and Basic (or suitable Basic Light) from Oracle. Match their versions and Windows architecture, then extract them into the same instantclient folder.

3. Confirm that sqlplus.exe and oci.dll are together. Install the Microsoft runtime required by that Instant Client release. Do not fetch individual DLLs from third-party sites or copy them into Windows system directories.

Keep the installation separate from your scripts

```
<INSTANT_CLIENT_DIR>\sqlplus.exe
<INSTANT_CLIENT_DIR>\oci.dll
```

Replace <INSTANT_CLIENT_DIR> with your actual extracted Instant Client folder. Do not type the angle brackets. The installation can be separate from the assessment package because PATH will locate the executable.

SQL*Plus does not require Java. No JDK installation or JAVA_HOME setup is needed for Part A. Any later Java-application assessment prerequisites belong to that separate application workflow.

Python is not required for customer collection. Optional host and pack-validation scripts have their own documented shell requirements.

2b. Windows: make the SQL*Plus PATH permanent

PATH lets Windows find sqlplus.exe while you remain in the assessment package folder. Configure your user PATH; do not change machine-wide settings without the desktop owner's approval.

1. Locate the actual Instant Client directory containing both sqlplus.exe and oci.dll.

2. Search Windows for 'Edit environment variables for your account'. Under User variables, select Path, choose Edit, then New.

3. Add the directory as one entry, without surrounding quotes. Replace the placeholder below with the real directory. Do not add a ZIP file, an executable filename or the SQL-script folder.

```
<INSTANT_CLIENT_DIR>
```

4. Preserve all existing PATH entries and save the dialogs. Close and reopen the terminal application; restart VS Code when using its integrated terminal. Another tab in an old application can retain the old environment.

Verify which executable Windows finds

```
Get-Command sqlplus.exe -All | Select-Object Source
sqlplus.exe -V
```

The resolved location must match the intended approved client and -V must print a version banner without a DLL/startup error. Multiple matches are listed in resolution order. Ask the desktop owner to resolve conflicting machine-level entries rather than deleting corporate installations.

Assignments such as $env:Path = ... are temporary, not permanent. Do not overwrite the entire PATH or replace it with a combined, possibly truncated setx value.

2c. Windows: verify SQL*Plus before collection

Open a fresh PowerShell window and change to the extracted package root, not the Instant Client installation directory. Replace the entire bracketed placeholder before execution:

```
Set-Location "C:\<PACKAGE_ROOT>"
Get-Command sqlplus.exe -All | Select-Object Source
sqlplus.exe -V
```

The version command must work from this package folder. PATH finds the executable; the working directory resolves relative @scripts paths. Do not change to the Oracle installation folder to launch the client.

```
sqlplus /nolog
```

This opens SQL*Plus without credentials in the process command line. Follow section 4 to connect securely, then run script 99 once as shown in section 5. Script 99 initializes output and creates its new Oracle run folders automatically.

Stop and correct workstation errors first

- 'Not recognized' or 'command not found': verify the executable directory, correct the user PATH, reopen the terminal application and check Get-Command again.
- Missing OCI.dll: merge the official matching Instant Client Basic and SQL*Plus packages and verify their required Microsoft runtime.
- Unable to open a script or SP2-0310: verify the package root, script filename and relative path. PATH is not a SQL-script search path.

CHECKPOINT A0 SQL*Plus resolves from the package root and prints its version without errors. Record the SQL*Plus client version. Resolve failures before section 4. Use script 99 after connecting; no separate script 00 is needed. These checks do not prove connectivity, privileges or source-version compatibility.

3. Prepare the package and evidence location

1. Obtain the complete runbook folder, not only the PDF, in an approved writable location. Keep instructions, scripts and output together, including all 99 helpers and scripts\validation\required-files.txt. Blank CSV record templates are in instructions. Source-code folders outside runbook are separate, not runtime dependencies.

2. Agree and protect the evidence location with the customer. Restrict its ACL/directory permissions before collection, keep it outside source control and unapproved synchronization, and confirm sufficient free space.

3. Script 99 creates the Oracle output folders automatically below the package's output directory. Each invocation gets a new UTC/GUID-based folder; do not rename or reuse a previous run folder. No manual per-owner folders or script 00 calls are needed for the guided run.

4. Start SQL*Plus from the package root. The following tree is illustrative: angle-bracketed names describe your actual location or automatically generated names, not text to type.

```
C:\<PACKAGE_ROOT>\
  instructions\
  scripts\oracle\
  scripts\os\
  scripts\perfstat\
  scripts\validation\
  scripts\post-assessment-report\
  output\<GENERATED_RUN_FOLDER>\
    oracle\database\
    oracle\schemas\<SELECTED_SCHEMA>\
    performance\window_01\
```

Optional host, Statspack and VS Code evidence use separately approved folders. They are not created or collected automatically by the Oracle SQL driver.

PLACEHOLDER RULE Anything inside <angle brackets> is a placeholder. Replace the entire token, including both brackets, with an approved value before running a command. Never run an example that still contains <...>. Keep an executable's path separate from your package working directory.

A PDB service identifies the database container; a schema owner identifies the application objects inside it; an assessment username identifies the login account. They are not interchangeable. Script 99 displays the actual owners and accepts their numeric menu selections.

4. Connect to the approved database or PDB

1. Complete CHECKPOINT A0. Open SQL*Plus in a terminal, not a SQL Developer worksheet. Change to the extracted package root, not an Oracle installation directory.

```
Set-Location "C:\<PACKAGE_ROOT>"
sqlplus /nolog
```

Use SQL*Plus throughout Oracle collection and recovery. Execute subsequent SQL commands at its SQL> prompt, not in PowerShell or a graphical worksheet.

2. At the SQL prompt, use the customer-approved connection method. Choose ONE of the following alternatives after replacing its placeholders. The first requires an existing approved wallet alias; the second prompts for a password without putting it in the command.

```
connect /@<APPROVED_PDB_OR_DATABASE_ALIAS>
-- OR, when approved for interactive authentication:
connect <ASSESSMENT_USER>@"//<DB_HOST>:<DB_PORT>/<APPROVED_SERVICE>"
```

Keep the double quotes around the full //host:port/service identifier. This avoids SQL*Plus parsing the service slash as part of another login option when the password is omitted. Enter the password only at the password prompt; do not add it to this command.

For a CDB application assessment, connect directly to the approved application PDB service. For a non-CDB/10g source, use the approved non-CDB service. Do not substitute the CDB SID for a PDB service. Keep the approved non-SYS assessment account and default role unless the DBA has authorized an alternative.

3. Confirm the host, service, environment and approved schema list with the DBA. Script 99 prints database/container evidence and asks you to confirm scope before schema selection. It does not reconnect, grant privileges or enumerate and collect every PDB automatically.

CHECKPOINT A1 You are authenticated to the intended service using the approved account. Continue to section 5. There is no separate script 00 setup step: the guided runner calls it for each output folder.

Keep a session transcript without recording credentials

Use approved terminal/session logging only after authentication. Each collector starts and stops its own SPOOL, so an outer SPOOL is not a master transcript. Review individual files for errors as well. The original collector scripts remain available for DBA-directed reruns, not the normal end-user sequence.

5. Run the complete guided Oracle collection

Run this ONE command at the connected SQL*Plus prompt from the package root. It contains no customer-specific placeholders:

```
@scripts\oracle\99_run_complete_assessment.sql
```

Script 99 now orchestrates every supplied numbered Oracle collector: 00-30 and 33. It collects the current database/container, every explicitly selected schema, and a start/end workload pair. Sections 6 and 7 explain steps already performed by this same invocation; do NOT run the old per-schema blocks again.

Answer the guided prompts

1. Choose the CLIENT operating system: 1 for Windows or 2 for Linux/macOS. This is the machine running SQL*Plus, not the database server.

2. Read the compact CURRENT ORACLE CONNECTION summary, shown before CSV collection. Confirm the Active container/PDB and Connected service. Login account and Session current schema identify the session, not the application owners. Enter 1 only for the approved container; 0 stops so you can reconnect.

3. Read the schema menu, which shows actual owner names and object counts. Enter how many approved owners you will select, then enter each owner's displayed NUMBER once. Do not enter a PDB name, username, placeholder, wildcard, or comma-separated expression.

4. Enter the positive workload interval in minutes that the DBA/application owner approved. When START is captured, the runner pauses while the representative workload runs.

5. The runner shows Timer START and Earliest END in UTC. The interval starts after the START snapshots, not when the whole assessment or interval-entry prompt began. Keep the session open and press Enter to check readiness.

An early check is NOT a fatal error. It displays the remaining seconds and returns to SQL>, still connected. The original START and all completed inventory are preserved. When ready, recheck with this command; do not rerun script 99:

```
@scripts\oracle\99_complete_workload.sql
```

If the connection was lost, reconnect to the original approved PDB and use the recovery file saved in that run folder. Replace the placeholder with the reported folder name:

```
@output\<GENERATED_RUN_FOLDER>\resume_workload.sql
```

Recovery verifies the original database/PDB and instance startup markers before capturing END. A changed target or restarted instance is blocked without touching snapshots. Repeating completion does not overwrite END files. Recovery captures END at the actual current time, so a delayed recovery produces a longer interval, not a recreated historical two-minute sample.

What remains separate

Optional host evidence and existing approved Statspack require different access and remain in sections 8-10. No AWR/ASH, source export, Statspack installation, Azure provisioning or VS Code conversion is authorized or executed by script 99.

CHECKPOINT A2 All numbered Oracle collectors have been invoked for the selected scope. The status file says VALIDATION_REQUIRED, not that every query passed. Keep the reported run folder and continue through sections 6, 7 and 11 before accepting the evidence.

6. Review the collected application schemas

AUTOMATIC IN SCRIPT 99 You do not need to reconnect, initialize script 00, or run scripts 08-17 and 33 one by one. The guided runner invokes that complete set for EACH owner you selected.

The schema menu is frozen for the run, so a changing database-user list cannot silently change your selection. Only numeric choices are accepted; actual owner names come from a filtered catalog. Duplicate or invalid choices stop explicitly instead of collecting the wrong schema. At workload decisions, the connection summary lists Collection owners separately from Session current schema.

Review each automatically created owner folder:

`output\<GENERATED_RUN_FOLDER>\oracle\schemas\<SELECTED_SCHEMA>\`

These outputs cover account/object counts, tables/indexes/partitions/LOBs, data types, source size, Oracle features, dependencies/links, jobs/queues/directories/external tables and privileges. The code collector records counts without exporting application source text.

The PDB service is NOT the schema owner. Choose the owners agreed with the DBA based on the displayed names and object counts. Do not select the assessment login merely because you use it to connect.

For another approved PDB, connect to that PDB and start a NEW script 99 run. The runner operates on its current connection; it does not switch containers or broaden permissions. If CDB-root evidence is separately required, agree a DBA-run root collection and keep it separate from application-PDB evidence.

Quoted or mixed-case owner names are not supported by these substitution-based collectors. The menu excludes them and Oracle-maintained accounts where the release exposes that flag. Such cases require an explicit DBA-reviewed manual collection, not ad-hoc escaping or wildcard scope.

CHECKPOINT A3 Every approved selected owner has its expected files in its own folder. Cross-check tables, indexes, partitions, invalid objects and dependencies with the DBA. Empty output from a wrong container or an owner with no application objects is not proof of completeness.

7. Complete the guided workload window

AUTOMATIC IN SCRIPT 99 The runner calls script 21, pauses for your agreed workload interval, and then calls scripts 22, 23 and 24. Do NOT run them a second time during a normal guided invocation.

Agree a representative normal, peak, batch or business-close period with the application owner. During the pause, let that real workload run and perform any separately approved, time-aligned host sampling. Do not generate artificial workload, restart the database or reconnect this collection session.

Press Enter to check elapsed time. If the agreed minimum has not elapsed, the runner shows remaining seconds and returns to the connected SQL> prompt without capturing END. Later run 99_complete_workload.sql as shown in section 5; the timer and START files are not reset. After a disconnect, use the saved per-run recovery file instead.

`output\<GENERATED_RUN_FOLDER>\performance\window_01\`

One invocation starts one workload window. Do not rerun the entire assessment to retry that unfinished window. Retain its original folder, workload label and actual UTC boundaries. A separate representative window needs a new START in a separate folder; do not overwrite or mix snapshots. Ask the DBA to approve that separate collection if needed.

Validate the pair before accepting it

- Match DBID, INST_ID, INSTANCE_NAME and STARTUP_TIME and require a positive interval. Discard restart-spanning, negative/reset or mismatched-instance intervals; the elapsed-time guard alone does not prove a valid counter pair.
- Both SYSSTAT and time-model start/end files are required. Top-SQL evidence contains identifiers and aggregate counters, not query text or bind values.
- Counter rates are (end - start) / elapsed seconds. Time-model values are microseconds; CPU used by this session is centiseconds; convert bytes to MiB using 1,048,576.
- Wait output is cumulative since startup and SYSMETRIC covers a short internal window. Neither automatically describes the entire start/end workload period.

CHECKPOINT A4 Every agreed workload period has a separate valid pair and label. Record missing or low-activity periods as evidence gaps, not production or peak requirements. Continue to the mandatory output-pack review in section 11.

8. Optional existing Statspack history

Optional Statspack uses its own sequence in scripts\perfstat: 01 lists snapshots; 02 collects metrics for DBA-selected IDs. These numbers are separate from scripts\oracle. Use only when the DBA confirms an existing approved Statspack owner/service. No installation, snapshot creation or purging is included. Existing history can provide useful business-period coverage.

List snapshots with SQL only

Create an existing protected output folder, open a fresh authenticated SQL*Plus session from C:\<PACKAGE_ROOT>, and run:

```
sqlplus /nolog
connect /@<APPROVED_ROOT_ALIAS>
@scripts\perfstat\01_list_perfstat_snapshots.sql C:\<PACKAGE_ROOT>\output\<RUN_ID>\statspack <PERFSTAT_OWNER>
```

SESSION ENDS The existing Statspack helpers terminate SQL*Plus on completion. Reopen the client and reconnect before the next helper. Do not assume output_dir, session settings or authentication persist.

The snapshot catalog helper expects visibility of STATS$SNAPSHOT, STATS$SYSSTAT, STATS$SYS_TIME_MODEL and STATS$SYSTEM_EVENT. If an older installation lacks one, record the limitation and obtain an approved DBA-generated report; do not install or modify Statspack to satisfy this runbook.

Select and collect a valid interval

1. Have the DBA choose begin/end snapshot IDs representing the same DBID, valid startup period and all required RAC instances. Confirm the snapshots' database time zone/offset.

2. Replace the illustrative 1000 and 1030 below. Use a fresh folder for each selected interval and keep the selection record.

```
sqlplus /nolog
connect /@<APPROVED_ROOT_ALIAS>
@scripts\perfstat\02_collect_perfstat_metrics.sql C:\<PACKAGE_ROOT>\output\<RUN_ID>\statspack <PERFSTAT_OWNER>
<BEGIN_SNAP_ID> <END_SNAP_ID>
```

Keep snapshot_catalog.csv, perfstat_metric_availability.csv, perfstat_metrics.csv and perfstat_top_waits.csv. Inspect availability/interval status and all ORA errors; do not calculate rates from invalid intervals.

LICENSING BOUNDARY If no approved Statspack history exists, retain the V$/GV$ snapshot path and record the historical gap. AWR/ASH are not an automatic substitute. The optional AWR SQL is outside this operator path and remains separately approval- and validation-gated.

9. Optional Linux Oracle-host evidence

Run on the actual approved Oracle host, not merely on an unrelated workstation. Obtain host-owner approval, agree sample count/interval, and create a new protected output directory for this host/window. Copy the supplied Linux collector unchanged using the approved transfer process.

```
bash ./collect_linux_host_info.sh <APPROVED_OUTPUT_ROOT>/os/<HOST_ALIAS>/<WORKLOAD_WINDOW> <SAMPLE_COUNT>
<SAMPLE_INTERVAL_SECONDS>
echo $?
```

Bash must already be available. The example collects five reports at one-second intervals per utility; it is a functional example, not a recommendation for the duration of a customer sizing study.

Output group	Review
OS, kernel, CPU, memory, uptime	Platform/capacity, current use and restart context.
Filesystems, block devices and LVM	Allocated/free space and physical/logical topology; do not double count layers.
Network, NUMA, hugepages, limits	Optional topology and configuration; shell ulimit is not the Oracle process limit.
vmstat / iostat / sar / mpstat	CPU, memory, I/O and latency observations; actual sampling windows are sequential.

Read host_collection_status.txt

- Samplers run sequentially: total runtime is not simply count x interval. The default four-tool sequence takes roughly 20 seconds plus inventory overhead.
- The first vmstat and iostat reports summarize since-boot activity. Exclude those reports from interval-rate analysis and use at least two reports.
- CAPTURED means the command succeeded; FAILED and diagnostics need investigation. LVM permission diagnostics are failures even if a tool returns zero.
- Exit 0 is COMPLETE_WITH_REVIEW; exit 1 is PARTIAL_FAILURE; exit 2 is invalid arguments. SKIPPED utilities are evidence gaps and are never installed automatically.
- Directories are protected as mode 700 and files use restrictive umask. Obtain separate approval before any sudo rerun; the script never elevates itself.

CHECKPOINT A5-LINUX Keep all TXT files and status log. Record failed/skipped tools, actual start/end/elapsed times and user. Do not sum physical-volume and logical-volume capacity as separate storage. Redact network/host details only in an approved transfer copy.

10. Optional Windows Oracle-host evidence

Run on the approved Oracle Windows host. Windows PowerShell 5.1 or PowerShell 7 is sufficient; no Python, new module, policy bypass or automatic elevation is required. Use an output folder with customer-approved inherited ACLs.

```
.\collect_windows_host_info.ps1 `
  -OutputDirectory <APPROVED_OUTPUT_ROOT>\os\<HOST_ALIAS>\<WORKLOAD_WINDOW> `
  -SampleCount <SAMPLE_COUNT> -SampleIntervalSeconds <SAMPLE_INTERVAL_SECONDS>
```

CSV	Grain and units
os_and_memory.csv	Host/OS; bytes and UTC boot time.
cpu_inventory.csv	Processor record; cores/threads; clock MHz.
disk_layout.csv / volume_capacity.csv	Physical disk or volume; size/free bytes; do not double count layers.
pagefile_usage.csv	Pagefile; sizes/usage in MiB.
network_interfaces.csv	Adapter inventory and reported link speed.
numa_inventory.csv	Per-node memory in MiB, excluding _Total; not complete CPU-affinity topology.
performance_samples.csv	Sample x counter; SampleNumber, TimestampUtc, Path, CookedValue, CounterStatus.

Counter sampling and validation

Nine counters are sampled together after inventory: total CPU %, available memory MiB, disk read/write bytes/s, read/write operations/s, read/write latency seconds, and disk queue requests. Three sample sets give 27 rows. Values use dot decimals regardless of regional formatting; timestamps are ISO 8601 UTC. PDH status 0 or 1 is valid.

- Review FAILED_COUNT, SKIPPED_COUNT, elapsed time and command timestamps. Missing cmdlets or empty results are explicit gaps; query/invalid-counter failures produce exit 1 and PARTIAL_FAILURE.
- No output ACL, system setting, installed module, execution policy or elevation is changed. Follow the customer's approved signing/execution procedure if policy blocks a script.
- English counter paths may fail on localized Windows; record the gap and ask the host owner for an approved mapping. Do not infer zero utilization.
- Host-wide PhysicalDisk(_Total) includes non-Oracle activity. Do not treat a local workstation smoke test as Oracle-host sizing evidence.

CHECKPOINT A5-WINDOWS Eight nonempty CSVs plus status log are expected when all providers are available. Investigate any skipped/failed provider. Inventory and initialization add runtime beyond the nominal sampling count/interval.

11. Validate and freeze the collection pack

1. Use Appendix A to check all database-wide, owner and window filenames. Compare every owner to the allow-list and every window to the planned business periods.

2. Inspect CSV/TXT outputs and transcripts for ORA-, SP2- and TNS- errors, line wrapping, unexpected headers, missing fields and zero-byte files. Missing rows may mean scope/privilege gaps rather than absence.

3. Confirm DBID/database/container/instance, UTC time, startup, script/package revision and operator. Record configured, measured, inferred and questionnaire values separately.

4. For files used in spreadsheets, import as text where appropriate to preserve identifiers/large numbers and avoid formula execution. Never execute content copied from evidence. Inspect quoting and delimiters: legacy SQL*Plus output is comma-separated text, not a general RFC CSV serializer.

5. Review secrets, personal data, internal endpoints and unapproved source content. Keep a controlled original and create an approved redacted transfer copy.

Optional validators - not a Python prerequisite

```
.\scripts\validation\validate_output_pack.ps1 `
  -InputPath C:\<PACKAGE_ROOT>\output\<RUN_ID>
```

```
bash ./scripts/validation/validate_output_pack.sh ./output/<RUN_ID>
```

These package validators search nested directories, detect missing required names, empty files, Oracle error markers and secret-like patterns without printing matched values, and create a SHA-256 inventory. Review false positives. They do not prove correct SQL results, parse every CSV shape, or guarantee every approved owner/window is present.

SCOPE OF THE CHECKLIST required-files.txt is for the Oracle base pack, not an OS-only or extension-only directory. <OWNER> matches at least one owner, so manual per-owner/PDB/window reconciliation is mandatory. An unsupported optional output needs an approved gap entry; do not create a fake file to pass validation.

Part A exit gate

- Source and host evidence is complete or gaps are accepted with owners and impact.
- All representative windows are labeled valid/invalid with reasons; no silent resets or double counting.
- Freeze a dated copy before starting conversion so later changes cannot overwrite the baseline.

11a. Optional: generate the Oracle HTML report

OPTIONAL STEP Skip this section if you only need the CSV evidence pack. HTML generation is not required for collection or handover, is not invoked by script 99, and does not replace the mandatory evidence review in section 11.

The bundled scripts\post-assessment-report\New-OracleAssessmentReport.ps1 builds an offline oracle_report.html from one existing run. It does not connect to Oracle or change CSVs. SQL*Plus remains the only supported Oracle collection client. No separate report ZIP or package checksum manifest is needed.

1. Finish collection/recovery and close the CSV files before reporting. A stopped partial run can also be reported, but missing or invalid evidence will be labelled. Do not generate while the collector is writing files.

2. Keep the complete runbook scripts tree together. The report shares scripts\validation\required-files.txt with the validators. This is a required-output checklist, not a checksum file. PowerShell 7 or later is needed only for this optional step; no Java, Python, downloaded modules or internet access is required.

3. Open PowerShell, NOT the SQL> prompt, in the runbook root containing instructions, scripts and output. Replace each entire bracketed placeholder below with its actual folder name:

```
Set-Location "C:\<PACKAGE_ROOT>"
pwsh -NoProfile -File .\scripts\post-assessment-report\New-OracleAssessmentReport.ps1 `
  -InputPath ".\output\<GENERATED_RUN_FOLDER>"
```

Select one generated run folder containing oracle and performance, not their parent. The default HTML is written inside that run folder. Open it in a browser and keep it with the source CSVs so the evidence links work.

4. Review the report status. Exit 0 means the report was written without detected review issues. Exit 2 means it was written but warnings/gaps need review; it is not an HTML-generation failure. Exit 1 means generation failed. A report failure does not invalidate or erase the existing Oracle evidence.

Charts use valid START/END pairs and actual elapsed times. Cumulative waits/cached SQL are labelled separately. Two snapshots provide averages, not a continuous timeline or proof of a short spike. The report is not a migration-readiness score or a production-sizing recommendation.

If generated, include the HTML and its checksum in the approved handover manifest and protect it as sensitive metadata. If skipped, omit it: oracle_report.html is not a required collection file. Never add dummy data or an empty HTML file to make a run appear complete.

12. Azure readiness and cost approval

Part B prepares a disposable assessment environment. An Azure administrator performs the approved provisioning. This document gives manual portal steps; writing or reading the document does not authorize deployment, role changes, restarts or paid use.

Record before creating anything	Required decision
Tenant / subscription / resource group	Confirm signed-in directory, subscription ID, owner and environment.
Region and residency	Approve Azure PostgreSQL Flex server and Foundry regions and model deployment type/data processing geography.
Budget and expiry	Estimate compute hours, storage/backup, model usage, optional VM and network charges; set cost owner and cleanup date.
Network route	Assessment machine must reach Oracle, PostgreSQL 5432, Foundry HTTPS, Marketplace/Copilot and required repository endpoints.
Identity	Authorized provisioning identity; dedicated conversion DB user; Foundry inference identity; no shared secrets in evidence.
Scratch version and capacity	PostgreSQL 15+; match planned production major version. Stable connection capacity, not production sizing.
Foundry model/quota	Current schema tutorial: gpt-5.2 and deployment quota >1,000,000 TPM [R1]. Confirm availability and subscription quota first.

Network decision

Prefer approved private connectivity and DNS: the workstation or approved assessment VM needs routing/VPN to each private endpoint. Private networking is not achieved just by choosing a private resource in the portal. If public access is explicitly approved, allow only required source egress IPs and retain TLS validation. Never open all IPs, expose Oracle publicly, or enable broad Azure-service access as a shortcut.

Cost and responsibility

The scratch server validates DDL; it does not need production data capacity. Choose an approved stable compute/storage configuration with the platform owner, not a final production SKU. Storage/backup and other resources may continue billing when compute is stopped; model TPM quota is not a spend limit. Review budgets/alerts and actual regional prices.

GATE G2/G3 Obtain budget, network, role-assignment and AI-processing approval. Stop if the required model/quota, source driver/version, data residency or network path is unavailable. Do not silently substitute another model or weaken security controls.

13. Create the Azure PostgreSQL Flex scratch server

Portal path checked against the official create-server quickstart [R5]. Prefer an existing dedicated assessment server if its owner confirms it is isolated and suitable. Do not reuse production.

1. Sign in at portal.azure.com. Confirm the correct directory/tenant and subscription. Open Create a resource > Databases > Azure Database for PostgreSQL flexible server > Create.

2. Under Basics, select the approved subscription and resource group, globally unique non-sensitive server name, region and PostgreSQL major version (15 or later and aligned with intended target).

3. Open Configure server. Select approved compute tier, size, storage and backup settings. Record choices/cost estimate. Do not treat quickstart defaults as a production sizing recommendation.

4. Choose the approved authentication option: Entra, PostgreSQL authentication or both. Have the administrator configure the authorized Entra/database identity or secure credential process. Keep credentials out of the runbook and screenshots.

5. Select the approved networking design. For private access, have the network owner complete routing and DNS from the assessment machine. For approved public access, restrict the firewall to actual egress IPs; do not allow 0.0.0.0/0 or broad service access.

6. Set encryption, backup, tags and lifecycle controls to customer policy. HA for this disposable scratch is a separate cost/risk decision, not a production architecture decision.

7. Review + create. Confirm G2 approval and the exact subscription/resource before selecting Create. Wait for deployment success; record resource ID, FQDN, version, region, compute/storage and owner.

8. Verify server status and permitted network reachability. The administrator prepares a dedicated database named <SCRATCH_DATABASE> (or approved equivalent) in the next section.

CHECKPOINT B1 A dedicated server is available, the client has a secure route to its FQDN on 5432, and authentication is approved. Do not use azure_sys, azure_maintenance, another team's database or a database with data to preserve.

Creation, networking and storage options change over time. Review the actual portal configuration and current official networking guidance before accepting it. A generic quickstart screenshot is not proof of your deployment's isolation.

14. Create the disposable database and access

The PostgreSQL administrator performs these steps using an approved SQL client or the PostgreSQL extension. Commands below mutate only the selected approved scratch environment; verify the active connection first. No production database is authorized.

Create a new database

```
-- Connected to the approved scratch SERVER, database postgres:
SELECT current_database(), current_user, version();
CREATE DATABASE <SCRATCH_DATABASE>;
```

Run CREATE DATABASE as a standalone statement, not inside an explicit transaction. If the name already exists, stop and have the owner confirm it is disposable or choose a new approved name. Do not drop it by copying a command from a tutorial.

Prepare the conversion identity

Have the database administrator authorize a dedicated conversion role with CONNECT and CREATE on <SCRATCH_DATABASE>, permission to create objects, and ownership/appropriate authority for matching schemas. CONNECT and database CREATE alone do not allow dropping another role's schemas. Extension installation can be done by an appropriately authorized administrator without giving the conversion identity that broader privilege [R1/R2].

Reconnect to the exact database

```
-- Connected to <SCRATCH_DATABASE>, not postgres or another database:
SELECT current_database(), current_user, version();
SELECT nspname FROM pg_namespace
 WHERE nspname NOT LIKE 'pg_%' AND nspname <> 'information_schema'
 ORDER BY nspname;
```

A new database contains standard schemas. Ensure no application schemas/data or dependencies need preserving. Review unexpected schemas with the owner; do not use a row-count or size threshold as proof it is safe to destroy.

DESTRUCTIVE OPERATION - G4 Conversion issues DROP SCHEMA ... CASCADE for schemas matching the selected Oracle owners, then recreates them. This removes objects/data and can remove dependent objects in other schemas. Only the owner may approve that acknowledgment. A scratch backup is not permission to reuse production.

Checkpoint B2

Record the exact server FQDN, database, role and major version. Keep scratch credentials in the approved secure connection store. Capture only non-secret connection metadata and the owner's disposable-database approval.

15. Prepare extensions in the scratch database

The extension wizard has a Verify Extensions check. Prepare the extensions that apply to the selected schemas with the administrator, using the current official best practices [R2]. Do not install every possible extension without need or approval.

Candidate	When needed
orafce	Oracle-compatible built-in functions/packages; check behavioral gaps.
uuid-ossp / pgcrypto	UUID or cryptographic functions referenced by converted objects.
tablefunc / fuzzystrmatch / pg_trgm	Crosstab, similarity, trigram/search transformations.
postgis and related components	Spatial/topology/geocoder objects only if applicable.
pg_partman	Approved partition maintenance design.
pg_stat_statements / plpgsql_check	Approved telemetry and additional PL/pgSQL static checks.
dblink	If generated autonomous-transaction logic requires it; review security/semantics.

1. Azure portal > selected scratch server > Server parameters. Find azure.extensions, add the approved required extensions and Save. Confirm support for the selected PostgreSQL major version.

2. Where required, configure shared_preload_libraries while preserving existing values. Current guidance calls out pg_stat_statements and plpgsql_check; pg_partman background maintenance uses pg_partman_bgw. Obtain restart approval for this dedicated server before applying a required restart.

3. Connect to <SCRATCH_DATABASE> as an authorized extension administrator. Execute only the approved CREATE EXTENSION statements, not the whole list by default.

```
SELECT current_database(), current_user;
-- Examples ONLY when approved, supported and allow-listed:
CREATE EXTENSION IF NOT EXISTS orafce;
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS plpgsql_check;
SELECT extname, extversion FROM pg_extension ORDER BY extname;
```

Review search_path for installed Oracle-compatible schemas using [R2]. Include only installed, trusted schemas; approve any ALTER DATABASE/ALTER ROLE setting and reconnect afterwards. Do not paste an all-extension search_path into a database with untrusted CREATE privileges.

CHECKPOINT B3 Required extensions are installed in the selected database, not merely server-allow-listed. Re-run Verify Extensions in the wizard. Record any intentionally missing extension and the affected object/validation limitation. Compilation/static checks are not runtime equivalence.

16. Create the Foundry deployment

Use an existing approved compatible deployment where possible. The current schema-conversion tutorial requires model gpt-5.2; the deployment can have a different name, for example assessment-gpt52. Do not confuse the deployment's user-defined name with its underlying model [R1].

1. Open ai.azure.com and sign in to the approved tenant. Use the current Foundry portal. Select the approved project/resource or Create new project, and review advanced options for subscription, resource group and location [R6].

2. Before creation, confirm resource/network/residency and budget approval. The portal can create the parent Foundry resource and related resources; review exactly what will be created. This assessment does not require provisioning an agent, AI Search, or extra data services.

3. Open Discover > Models in the current portal, search for gpt-5.2 and open its model card. Confirm supported region, deployment type, model version and allowed usage. If not available, stop and resolve the requirement rather than substituting a model.

4. Select Deploy > Custom settings. Choose the approved deployment name and type. Review data-processing geography: a resource region alone does not guarantee inference stays there for every deployment type.

5. Allocate approved capacity. The schema tutorial currently instructs deployment quota greater than 1,000,000 tokens per minute; 2.3M TPM is its example, not a universal quota or price. Obtain required quota and cost approval before deployment.

6. Select Deploy only with G2/G3 approval. Wait for Succeeded. Under Build > Models, open the deployment details and record model name/version, deployment name/type, resource and region [R7].

7. Use a non-sensitive test prompt in the playground if permitted; verify authentication and deployment availability without pasting customer code.

CAPACITY GATE If your subscription cannot allocate the documented TPM, log the blocker and confirm supported operating guidance with Microsoft/customer platform owners. Do not reduce quota and describe the setup as fully matching the tutorial. TPM is a rate allocation, not an expenditure cap.

Record for the extension

Resource endpoint (not a project URL), deployment name, model/version, authentication method, deployment type, region, quota, approved identity and non-secret resource ID. Never include API keys or access tokens.

17. Foundry authentication and connectivity

Choice	Configuration and control
Microsoft Entra ID - preferred when supported	Sign in using the Azure Account extension/flow documented by the wizard. Authorized identity needs Foundry User on the Foundry resource. Have the RBAC owner grant only approved scope; resource creation permission is not inference permission.
API key - only if approved	Use the Foundry resource endpoint and key in the wizard's secure credential field. Obtain through the customer's secret-handling process; never paste into Word, scripts, project files or logs.

Endpoint format

The current tutorial accepts the Foundry resource URL in any of these resource-host formats. Copy the actual endpoint from the approved resource; do not construct a value from a project name or copy the resource's project /api/projects path.

```
https://<resource>.services.ai.azure.com
https://<resource>.openai.azure.com
https://<resource>.cognitiveservices.azure.com
```

1. Verify the assessment workstation/VM can resolve and reach the configured Foundry endpoint over HTTPS. A private endpoint requires approved DNS and route/VPN access from that machine.

2. Verify tenant, identity and role scope. Do not assume the Azure account browsing resources is necessarily the same identity selected for database authentication or Foundry inference.

3. Use Test Microsoft Foundry Connection in the migration wizard after entering endpoint/deployment/authentication (section 20). Preserve sanitized test outcome and errors, not the key/token.

4. If a 403/private-access or timeout occurs, check resource network policy, DNS and routing with the platform owner. Do not enable public access, disable TLS verification or expand roles as an unapproved workaround.

Checkpoint B4 - environment ready

- PostgreSQL scratch database is isolated, disposable, reachable and extension-ready.
- Foundry deployment is the correct model/version with approved quota, geography and inference access.
- AI processing of the selected definitions and any application code is authorized.
- Creation, any restart and role-assignment changes are recorded with owners/costs/expiry.

18. Install and connect the VS Code extension

Workstation prerequisites

Item	Current documented requirement [R1/R3]
VS Code	1.95.2 or later; record actual installed version.
PostgreSQL extension	Microsoft publisher; ID ms-ossdata.vscode-pgsql.
Schema-workflow OS	Windows x64, Linux x64, macOS 13+; check current compatibility.
GitHub Copilot	Documented Pro+, Business or Enterprise entitlement and customer policy approval.
Oracle driver	Thin by default; automatic thick-mode detection from Oracle network settings. No manual mode selector.
Network	Oracle, scratch PostgreSQL, Foundry, Marketplace/Copilot services and github.com/microsoft/pgsql-tools/.

1. In VS Code, open Extensions (Ctrl+Shift+X), search PostgreSQL, verify Microsoft publisher and the exact extension ID, then install/update per policy. Record the version.

2. Sign in to the approved Copilot account. For Entra-based Foundry authentication, complete the current Azure Account sign-in flow required by the tutorial. Do not change hidden pgsql.enableMigrations settings based on old instructions.

3. In the PostgreSQL extension panel, create a connection to the dedicated Flexible Server using its FQDN, port 5432, <SCRATCH_DATABASE> and the approved conversion identity. Use the supported password or Entra flow and preserve TLS verification.

4. Test and save the connection through the extension's secure mechanism. Open a SQL query and run SELECT current_database(), current_user, version(); confirm exact target.

5. Create an empty local folder for the assessment migration workspace and open it with File > Open Folder / a new VS Code window. Keep it separate from immutable source evidence and unrelated repositories.

OLDER ORACLE / THICK MODE The wizard's default thin driver does not establish support for an Oracle 10g source. Validate exact source-release and driver support first. If approved native-network settings require thick mode, install the compatible Oracle Instant Client through customer policy and set its path/configuration before launching VS Code. Never disable source encryption to force a connection.

Record relevant sqlnet.ora/tnsnames.ora configuration locations without copying secrets. Current documentation describes automatic mode selection; record the observed behavior and driver/client version when troubleshooting.

19. Create the project and review Discovery

Follow the documented migration wizard labels [R1]. Labels can change; if your build differs, use the current official tutorial and record the difference rather than guessing.

1. Open PostgreSQL > Migrations > Create Migration Project.

2. Enter a non-sensitive project name in the Migration Wizard and select Next.

3. On Connect to Oracle, enter approved host, port, database/service, username and password in secure fields. This source can be customer Oracle on-premises or in cloud; the CSVs are not a substitute for the live metadata connection.

4. Select Load Schemas. Confirm connection/service/container and choose only approved application owners. Do not choose all user schemas just because they are visible.

5. Select Next: Discovery and wait. Discovery inventories scope; no DDL is extracted at this stage.

6. Under Source permissions, inspect each result. Ask the DBA to resolve approved access issues, then select Refresh Discovery. Never grant privileges from an AI suggestion without approval.

7. Review Discovered, To be extracted, Excluded and Invalid objects. Expand supported and unsupported categories; compare with Part A object counts using the same container/owners and collection timing.

8. Under Dependent schemas, use Show diagram. Edge numbers represent dependency references, not extra object counts. Agree whether to add a missing application dependency; Oracle system-schema references are not additional migration scope.

9. If scope changes, use Previous to return to Connect to Oracle, update Schemas and repeat Discovery. Record exclusions and reasons; discovered count is not converted count.

10. Only after review, select I have reviewed the objects, exclusions and dependent schemas, then Next: PostgreSQL Connection.

CHECKPOINT C1 Save sanitized Discovery counts/screenshots and the owner/dependency decision. Reconcile differences rather than forcing SQL and extension counts to match: supported types, exclusions, invalids, partitions and scope can have different counting rules.

Do not proceed if

- Wrong source/container/owner, unapproved schema additions or unexplained metadata-access failure.
- Customer has not approved sending definitions to the model service.
- You cannot identify which scratch database will be affected by the next stage.

20. Configure scratch and model in the wizard

PostgreSQL Connection

1. Choose the saved Azure Database for PostgreSQL connection and select the dedicated <SCRATCH_DATABASE> database. Reconfirm server, role and PostgreSQL major version.

2. Select Verify Extensions. Compare warnings to the approved extension list. Ask the administrator to allow/install applicable missing extensions in this exact database; then select Verify Extensions again.

3. If a warning is intentionally left unresolved, record which objects/validations are affected and get an explicit decision. Continuing with a warning does not mean the requirement was met.

4. Read the acknowledgment: scratch schemas matching selected Oracle schemas are dropped and recreated, deleting existing objects/data, including possible cascade dependencies. Obtain G4 approval before checking it.

5. Select Next: Microsoft Foundry Model Configuration.

Microsoft Foundry Model Configuration

1. Enter the Foundry resource endpoint from section 17 and the deployment name from section 16. The actual model must be gpt-5.2 even if the deployment name is different.

2. Choose Microsoft Entra ID or API key according to the approved authentication design. For Entra, sign in with the documented Azure Account flow and verify Foundry User on the resource. For API key, enter it only in the intended secure field.

3. Select Test Microsoft Foundry Connection. Resolve failures with the platform owner; keep a sanitized outcome and the tested configuration/version.

4. After success, select Create Migration Project. Check project location and ensure no connection secrets were written into files approved for handover.

CHECKPOINT C2 Project created with the correct source owners, dedicated disposable scratch database, required extensions and tested Foundry deployment. Record G3/G4 approvals, model/deployment version and quota, extension version and UTC start time.

Before starting conversion

Ensure the workstation/approved VM can stay connected and avoid sleep/automatic shutdown during the approved conversion period. Record any long-run host policy changes separately; do not change them automatically. Consider an approved Azure assessment VM for large/long conversions, but treat it as additional paid infrastructure requiring approval.

21. Run and inspect schema conversion

1. On the main Migration Wizard, select Migrate. Monitor extraction/conversion/validation progress and logs. Do not point it to production or add extra source scope while a run is active.

2. When Schema conversion is complete appears, open Review conversion summary and preserve the HTML Conversion report. Review discovery, extraction and conversion stages separately.

3. Open Conversion object explorer. For both Converted and Not converted objects, compare Oracle/PostgreSQL definitions, dependencies, statuses and object notes.

4. Inspect routines for placeholders, raised errors, missing logic and unsupported features. A Converted status can include best-effort or placeholder output and does not guarantee compilation, complete implementation or business equivalence [R4].

5. Use Ask for explanation and Fix only within approved copied/generated PostgreSQL files. Review each change with the owner; record manual remediation, effort, risks and validation needed.

6. Keep generated object files, mappings, Coding Notes, reports, setup files and the complete project. After edits, separately verify object SQL and deploy.sql: editing one does not guarantee the other changed.

Stored database code versus application code

Oracle PL/SQL procedures/functions/packages/triggers are database definitions in the schema workflow. Application code (for example Java/C# data-access calls and embedded SQL) is a separate workflow in section 23. A source-size CSV is not a code conversion assessment.

Setup instruction - review, do not blindly execute

From Conversion summary, select Setup instruction when available. Read setup_instructions.md; opening it does not deploy anything. To validate a deployed schema, obtain separate approval and use a separate nonproduction destination database, not the disposable conversion scratch. The destination must contain no data you need to preserve if generated setup is destructive.

CHECKPOINT C3 Preserve full object-level results and unresolved issues, not only headline success percentage. Record exclusions, invalid source objects, failed/placeholder conversions, missing extensions and manual edits. Retain exact source definition/package/model versions necessary to reproduce the run.

22. Approve a separate validation destination

This section is an optional assessment follow-on needed when application conversion requires deployed-schema context. It is not a production rollout, a row-data migration or permission to run generated deployment scripts blindly.

1. Application and database owners approve a separate empty nonproduction validation database, its cost and deployment authority. Do not use the conversion scratch as a functional-test environment.

2. Read generated setup_instructions.md and deployment SQL in full. Confirm destination FQDN/database, extensions, search_path, role ownership and any destructive statements. Use the Setup Experience guidance [R8].

3. Deploy only with approval, using the current Setup Experience and readiness checks. Keep deployment results, failed/omitted objects, generated deploy.sql and the actual executed revision.

4. If per-object fixes are made afterwards, verify active editor connection and execute only complete, reviewed statements in the authorized nonproduction database. File edits, deployed-object updates and deploy.sql updates are three separate actions.

5. Use only approved synthetic/masked representative test data through a separate test-data process. This runbook does not move production rows to create validation data.

Validation	Evidence needed
Structural	Constraints, types, dependencies, indexes, object compile/deploy errors.
Semantics	Transactions, exceptions, null/empty strings, numeric precision, date/time, ordering and sequences.
Functional	Business workflow results checked independently of the converting AI model.
Performance / resilience	Separate representative workload, capacity, backup/restore, failover and concurrency tests.

NO SELF-CERTIFICATION Conversion and AI-generated validation can repeat the same mistake. Independent business and technical verification is required before production. Record untested items clearly; do not upgrade an assessment to a production-ready claim.

Checkpoint C4

Application-conversion context points to the authorized database with the converted schema actually deployed. If deployment is not approved, defer application conversion that requires this context and record the blocked scope.

23. Assess application code - preview workflow

This is optional and separately approved. Current Microsoft documentation describes application conversion as preview and uses GitHub Copilot Agent Mode, not the schema converter's Foundry deployment [R3]. A schema-only assessment does not require copying an application repository.

Prepare the approved code copy

1. Confirm application owner and AI-processing approval, repository scope, supported language/toolchain, Copilot entitlement and allowed Agent Mode tools.

2. Keep an immutable original/backup outside the migration project. Copy only approved code; exclude credentials, .env files, private keys, production exports, unrelated repositories and generated build caches.

3. Open the existing schema-conversion workspace. In Copilot Chat select Agent Mode and the model specified by the current application tutorial: Claude Sonnet 4.6 or Claude Opus 4.6 (or a currently documented higher supported option). This does not replace the gpt-5.2 schema requirement.

4. Copy the selected codebase into the project's application_code folder. Record source version/commit and files included. Preserve Coding Notes from schema conversion to provide mapping/transformation context.

```
<workspace>\
  .github\postgres-migration\<project_name>\
    application_code\
      <approved_codebase>\
```

Run the application assessment

1. In the extension Migration panel select Migrate Application. On the form choose the approved copied application folder.

2. Select the nonproduction PostgreSQL database containing the deployed converted schema (section 22), or another authorized existing application-context database.

3. Select Convert Application. Monitor the generated Agent Mode TODO/tasks and tool actions as it converts files.

4. Review requests before approval. Do not allow arbitrary network uploads, production queries, credential reads, new paid resources or uncontrolled terminal commands.

CHECKPOINT C5 Only the copied approved code is modified. Original code is preserved, model/tool versions and source revision are recorded, and the database context is authorized. No application deployment or production row movement is performed.

24. Review application findings and changes

1. When conversion completes, preserve the generated report, task outcomes, TODOs, warnings, failed/unchanged files and recommended next steps.

2. Right-click a converted file and select Compare App Migration File Pairs where available. Review original versus converted logic, not only text changes.

3. Check connection/driver changes, SQL syntax, parameter binding, transaction boundaries, error handling, types/nulls, time zones and stored-routine call mappings.

4. Run only approved build/unit/integration tests against the authorized nonproduction environment with approved test data. Record exact commands, tool versions and results. Do not install missing toolchains without approval.

5. For Java, the tutorial describes an optional GitHub Copilot App Modernization extension integration. Install only if approved; auto-generated tests/compilation do not prove business equivalence.

6. Record each remaining issue with component/file, reason, business impact, owner, remediation estimate, proposed test and acceptance authority.

Keep	Why
Original copy and exact revision	Reproducible comparison and rollback of local assessment edits.
Converted code / diffs	Actual changes rather than only summarized claims.
Report / TODOs / errors	Unconverted scope, manual effort and residual risks.
Coding Notes / mapping context	Relate database changes to application assumptions.
Independent test results	Distinguish conversion suggestions from verified behavior.

ASSESSMENT BOUNDARY No production application deployment, connection-string cutover or live data movement is included. Compile success is not runtime equivalence. If code or tests were unavailable, mark that portion not assessed rather than inventing a completion percentage.

Checkpoint C6

The application owner accepts the assessment findings and untested scope. Preserve full evidence; redact secrets only on a controlled copy. Return original/converted source only through the approved IP-handling process.

25. Package, protect and hand over

1. Freeze the final evidence tree: Oracle/host collection, scope record, Azure non-secret configuration, full migration project, reports, original/converted code when authorized, and issue log.

2. Copy only approved project content into output\<RUN_ID>\vscode-assessment. Preserve the complete generated project structure and necessary hidden .github folder; do not rely on a single HTML report or assume a generic Export button exists in every extension version.

3. Inspect credentials, tokens, endpoints, personal data, source IP and unapproved content. Restrict a controlled original, then redact an approved transfer copy. Do not remove failed findings to make the assessment look successful.

4. Create a manifest: relative filename, bytes, SHA-256, collection UTC, database/container/instance/host alias, script/tool version, redaction status and notes. Hash after the final redaction, not before.

5. Create an archive with the approved tool; ordinary ZIP/tar is NOT encryption. Encrypt through the approved process and exchange the key through a different channel.

6. Transfer through approved secure exchange. Recipient verifies archive hash, opens it, checks completeness and confirms receipt/retention/cleanup instructions.

```
# Optional Windows checksum command (PowerShell, not SQL):
Get-FileHash -Algorithm SHA256 -LiteralPath "C:\<PACKAGE_ROOT>\<EVIDENCE_ARCHIVE>.zip"
```

Final operator checklist

- [ ] Every database/PDB/owner and workload window is accounted for; gaps have impact and owner.
- [ ] SQL/host errors, resets, skipped tools, unsupported versions and invalid intervals are visible.
- [ ] Discovery/extraction/conversion/deployment/application outcomes are distinguished.
- [ ] Full reports, object SQL, mappings, Coding Notes, setup files and intended deployment SQL retained.
- [ ] Secrets/IP sensitivity review and redaction approval complete; final manifest and archive hash match.
- [ ] Azure cost/resource owner, expiry and cleanup approval recorded; no destructive cleanup is assumed.
- [ ] Recipient confirmation and assessment lead sign-off recorded.

CLEANUP IS A SEPARATE APPROVAL Do not delete source data, production resources, shared resource groups or retained evidence. After acceptance, ask the customer to authorize cleanup of specifically identified temporary resources/copies. Stopping a server is not deleting all charges; verify remaining storage/backup/model resource costs.

26. Troubleshoot without weakening controls

Symptom	Safe action
ORA-00942 / ORA-01031	DBA checks version, container and approved view grants; record gap. Never self-grant.
ORA-00904 / missing counter	Version-specific column/view issue. Preserve error, obtain tested alternative; do not substitute zero.
ORA-12154 / 12514 / 12541	Check alias/service/listener, route and approved network access. Do not expose Oracle publicly.
Empty owner output	Verify approved owner, PDB/non-CDB and dictionary visibility; reconcile inventory.
SPOOL / path failure	Create the existing client-side directory and check approved permissions/no spaces.
Root PDB list empty / ASM no rows	Ask DBA for topology and scoped evidence; do not infer no PDB/ASM usage.
RAC / snapshot reset	Match identities/startup and aligned times; discard invalid interval and recollect.
Linux LVM permission error	Retain diagnostics; ask for separately approved elevated/DBA-run collection. No auto sudo.
Windows policy / localized counter	Use approved signed execution procedure or localized mapping; no policy bypass.
Migration feature absent	Verify Microsoft extension/version and current tutorial; do not invent a hidden setting.
Oracle thin/thick connection fails	Check supported source/client versions and approved Oracle network config; no encryption downgrade.
Scratch TLS / DNS / auth failure	Check FQDN/database/role, firewall/private route, DNS and certificate trust; no TLS bypass.
Verify Extensions warning	Admin allow-lists AND installs required extensions in selected database; record omissions.
Foundry 401/403 / quota / 429	Check endpoint, tenant, deployment/model, Foundry User scope, network and documented TPM; do not swap model silently.
Conversion timeout / partial results	Preserve logs/project and scope counts; retry only per current guidance, not destructive guessing.
Converted but unusable object	Review placeholders/notes/dependencies; record remediation and independent tests.
Object edited but deployment differs	Reconcile per-object SQL, deployed object and deploy.sql separately.
Secret found	Stop sharing, restrict access, notify owner; redact a copy, rehash and revalidate.

Issue record minimum: UTC time, step, tool/source version, non-secret error, affected scope, impact, owner, approved workaround and decision deadline.

27. Handover for report and rollout planning

The operator supplies evidence; the migration architect performs the offline analysis and prepares a customer-specific report and rollout plan. Do not size Flexible Server from allocated Oracle memory, host CPU count or a single maximum alone.

Report section	Required evidence and decision
Scope and confidence	Source version/topology, owners, periods, approvals, exclusions and evidence quality.
Capacity envelope	CPU demand, memory working set, IOPS/throughput/latency, connections, growth, temp/spill and redo/change rate; units and assumptions.
Schema and code	Discovery/extraction/conversion scope, manual/placeholder objects, application diffs and remediation estimates.
Azure candidates	Regional compute/storage/network/HA/backup/extension options, quota, pricing date and cost ranges.
Data migration design	Separate method for rows/LOBs, initial load, supported CDC if needed, downtime and reconciliation.
Rollout gates	Prototype, remediation, data-movement rehearsal, performance/restore/failover/security tests, acceptance, cutover and rollback.

Separate measurements from assumptions

- For each valid pair calculate delta and actual elapsed rate with documented units. RAC aggregation requires aligned per-instance intervals, not indiscriminate summing.
- Percentiles require sufficient representative observations. Do not compute a meaningful P95 from a three-sample smoke test.
- Oracle table NUM_ROWS/statistics and segments are estimates/allocation, not exact live row counts or PostgreSQL storage forecasts.
- Record expansion/growth/headroom and HA/backup costs separately. Prototype the candidate target with representative workload before production recommendation.

DATA MIGRATION REMAINS SEPARATE Schema conversion does not transfer application rows. The rollout plan must select a supported source-version method for bulk load/CDC if required, consistency, replay/retry, final synchronization, row/business-aggregate reconciliation and rollback. This runbook does not prescribe an unvalidated universal migration tool.

Assessment acceptance

Assessment lead: _____ Customer owner: _____
Date (UTC): _____ Evidence location/hash: _____
Accepted gaps and follow-up owners: _____

Appendix A. Expected collection files

This is the supplied required-file checklist as of this document build. Replace <OWNER> with each approved owner. All filenames are relative to the appropriate database/owner/window output folder, not all at the evidence root. Optional unavailable features require a documented gap, not a fabricated file.

Database-wide folder: oracle\database
run_environment.csv
01_assessment_precheck.csv
02_database_identity.csv
02_version_banners.csv
03_instance_configuration.csv
03_host_os_stats.csv
04_cdb_pdb_inventory.csv
05_initialization_parameters.csv
06_character_set_nls.csv
07_tablespace_capacity.csv
07_tempfile_capacity.csv
18_session_limits.csv
18_session_profile.csv
19_sga_pga_memory.csv
20_io_redo_counters.csv
25_rac_inventory.csv
26_asm_inventory.csv
27_data_guard_inventory.csv
28_growth_capacity.csv
29_backup_recovery.csv
29_rman_backup_jobs.csv
30_awr_ash_licensing_checkpoint.csv
99_run_complete_assessment.txt

Script 04 can also be retained per selected PDB to prove active-container scope. Do not merge PDB/root copies into one file or sum duplicate DB-wide capacity.

Appendix A continued - owner and window files

Every approved owner/PDB folder
08_schema_inventory_<OWNER>.csv
09_object_type_inventory_<OWNER>.csv
10_table_index_partition_<OWNER>.csv
11_lob_inventory_<OWNER>.csv
12_data_type_compatibility_<OWNER>.csv
13_code_object_inventory_<OWNER>.csv
14_feature_usage_<OWNER>.csv
15_dependencies_<OWNER>.csv
15_database_links.csv
16_jobs_queues_directories_<OWNER>.csv
17_security_summary_<OWNER>.csv
33_configuration_risks_<OWNER>.csv

Every workload-window folder
run_environment.csv
21_workload_start.csv
21_workload_time_model_start.csv
22_workload_end.csv
22_workload_time_model_end.csv
23_top_sql_metadata.csv
24_wait_events.csv
24_system_metrics.csv

Host output names are described in sections 9/10. Each host run also needs host_collection_status.txt. Existing Statspack history uses its own snapshot/availability/metric/wait files (section 8). No base output pack should contain source-export or AWR files without separate approval.

Appendix B. Operator working record

Complete these fields for each customer engagement. Use additional copies for multiple databases, owners or measurement windows. Keep passwords and API keys out of every field.

Field	Customer value / evidence reference
Customer / engagement / classification	_____
Run ID / package revision or SHA-256	_____
Operator / assessment lead / contacts	_____
DB unique name / DBID / environment	_____
Oracle version / patch / edition	_____
Service / root-PDB or non-CDB / RAC instances	_____
Host aliases / OS / owner allow-list	_____
Window label / UTC start / end / workload	_____
Restart/reset check / validity / gaps	_____
G1 source / host sampling approval	_____
Tenant / subscription / RG / region	_____
Scratch FQDN / database / PG major / role	_____
Foundry endpoint / deployment / model/version	_____
Deployment type / region / TPM / auth identity	_____
G2 budget / roles / network approval	_____
G3 AI/IP and G4 scratch-destruction approval	_____
VS Code / extension / Copilot model versions	_____
G5 approved code path / source revision	_____
Reports / project path / open issues	_____
Archive / SHA-256 / receipt / expiry / cleanup owner	_____

Appendix C. Validation status and references

What was actually tested

Component	Evidence / limits
Oracle base collection	Oracle 21.3 / SQL*Plus 21.3; root inventory, HR/CO/SH owner inventory, valid 66-second smoke pair; 76 CSVs and catalog reconciliation. Not 10g/11g/12c/19c or RAC/ASM/Data Guard certification.
Linux collector	Oracle Linux 7.9 lab, normal-user permission failures and approved sudo run; 18 available outputs, four samplers; NUMA/multipath tools absent.
Windows collector	Windows 11 workstation without elevation; PowerShell 7.6.7 and Windows PowerShell 5.1; eight CSVs, 27 counter rows per run.
Azure and extension procedure	Documentation verified on 20 October 2026, not a claim that a new Azure/Foundry or end-to-end conversion deployment was executed for this document.

These were functionality/smoke tests, not representative customer workload or a production sizing proof. Matching-version, topology, network and privilege validation remains mandatory before customer use.

Guided runner 1.2 was smoke-tested with Oracle 21c SQL*Plus using administrative access to the lab PDB and the HR/SH sample owners. All 58 expected CSV outputs were nonempty. Early checks returned to the connected prompt with the original inventory/START files unchanged. Saved recovery was exercised after reconnect; wrong-container and changed startup markers were blocked, and repeated completion preserved END. Windows folder creation was checked separately. This does not certify every customer account, client or Oracle release, and the short lab interval is not sizing evidence. The quoted Easy Connect identifier was separately verified to reach the password prompt in Windows SQL*Plus 23.26 without supplying a password or attempting database authentication. Connection checkpoints are narrow, labelled displays; raw CSV collectors retain their full field widths and visible errors. Do not infer identity from the wrapped CSV output, and do not enable TERMOUT OFF to hide it because that also hides Oracle errors.

Official sources - recheck before execution

R1 - Schema conversion tutorial (wizard, prerequisites, privileges, quota)
https://learn.microsoft.com/en-us/azure/postgresql/migrate/oracle-conversions-schema/schema-conversions-tutorial

R2 - Schema conversion best practices (scratch, extensions, search_path)
https://learn.microsoft.com/en-us/azure/postgresql/migrate/oracle-conversions-schema/schema-conversions-best-practices

R3 - Application conversion tutorial (preview, model, copy, workflow)
https://learn.microsoft.com/en-us/azure/postgresql/migrate/oracle-conversions-application/app-conversions-tutorial

R4 - Schema conversion limitations (placeholders, support boundaries)
https://learn.microsoft.com/en-us/azure/postgresql/migrate/oracle-conversions-schema/schema-conversions-limitations

Appendix C continued - Azure and product sources

R5 - Create a Flexible Server (portal quickstart)
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/quickstart-create-server

R6 - Create a Foundry project
https://learn.microsoft.com/en-us/azure/foundry/how-to/create-projects

R7 - Deploy models in the Foundry portal
https://learn.microsoft.com/en-us/azure/foundry/foundry-models/how-to/deploy-foundry-models

R8 - Schema Setup Experience
https://learn.microsoft.com/en-us/azure/postgresql/migrate/oracle-conversions-schema/schema-conversions-setup

R9 - PostgreSQL extension connections and identity
https://learn.microsoft.com/en-us/azure/postgresql/development/vs-code-extension/connections

R10 - Flexible Server private networking
https://learn.microsoft.com/en-us/azure/postgresql/network/concepts-networking-private

R11 - Foundry role-based access control
https://learn.microsoft.com/en-us/azure/foundry/concepts/rbac-foundry

R12 - Extension availability by PostgreSQL version
https://learn.microsoft.com/en-us/azure/postgresql/extensions/concepts-extensions-by-engine

R13 - Oracle 10g ALL_SOURCE / DBA_SOURCE reference
https://docs.oracle.com/cd/B19306_01/server.102/b14237/statviews_2063.htm

R14 - Oracle licensing information (use release/contract-specific advice)
https://docs.oracle.com/en/database/oracle/oracle-database/19/dblic/Licensing-Information.html

This guide paraphrases operational instructions and links to the first-party sources; portal screenshots and customer settings are not reproduced. Current product documentation and customer policy take precedence if behavior changes.